

Bayesian Modeling of Market Volatility Following Political Communication Events

Shailesh Bolduc, Jack Shi, Joyce Zhang, Eric Lee, Wesley Wang

Introduction

Those in positions of great influence have the ability to affect how people conduct themselves in regard to various situations. It is arguable that the one with the most influence in the United States of America is the President, Donald J. Trump at the time of writing. As such, various studies have sought to quantify these figures' influence on people, markets, goods, and more. One such study, for example, found that the market responded in a statistically significant manner to tweets from presidential candidates since 2016 that had positive linguistic tones. (Marinč et al.)

In this study, we seek to extend this area of research by investigating how political communication events influence short-term market volatility. The focus is on modeling how markets respond to new political information presented in Donald J. Trump's speeches and other recorded official political communications in his second term as president.

The project will analyze whether the topics of each political communication are associated with changes in market volatility. Uncertainty is measured using the VIX volatility index, which reflects expected volatility in the U.S. stock market (CBOE).

The objective is not to estimate a single fixed relationship between political communication and market volatility, but rather to estimate the posterior distribution of that relationship. Bayesian inference treats the parameters as distributions and observed data (political communications) is then used to update beliefs and obtain the posterior.

Data and Methodology

Model

To operationalize the proposed input-output Hidden Markov Model, we will leverage an existing open-source implementation of IOHMM. This framework provides a

flexible structure for modeling state transitions as functions of exogenous inputs, aligning well with our objective.

To make the model more concrete, we explicitly specify both the transition and emission components. For the transition model, we use a multinomial logistic (softmax) function to map NLP-derived features into state transition probabilities. Specifically, the probability of transitioning from state i to state j at time t is modeled as a function of X_t , where X_t includes topic scores from Seeded Latent Dirichlet Allocation. This allows us to quantify how different aspects of political communication shift the likelihood of entering higher-volatility regimes.

For the emission component, we assume that the observed market variable, namely changes in the VIX, follows a state-dependent distribution. A natural choice is a Gaussian distribution with state-specific mean and variance, which captures the idea that higher-volatility regimes are associated with both larger average movements and greater dispersion. We adopted an unsupervised learning approach in which the model was allowed to infer latent states when the number, but not criteria, for the labels are available. This allowed the model to discover regimes and patterns on its own, rather than rely on strict bounds that may not reflect the structure of the data.

The transition probabilities are parameterized as functions of NLP-derived features from political speeches, enabling the model to capture how linguistic signals influence regime dynamics. The emission component links latent states to observed market variables, specifically VIX movements. As a baseline, model estimation was conducted using multinomial logistic regression for point estimates of parameters. As an extension, estimation is then conducted using probabilistic inference techniques such as Baum-Welch Expectation-Maximization and Hamiltonian Monte Carlo No U-Turn Sampling (NUTS), allowing for robust estimation of both latent states and model parameters.

Data

Our empirical analysis integrates financial market data with textual data from political communications. We will use daily observations of the CBOE Volatility Index (VIX) as a proxy for market uncertainty (CBOE). The dataset spans May 2025 to May 2026 and is transformed into a time series of volatility changes to reflect regime shifts. We also include

market-level control variables, such as S&P 500 return, using publicly available data from Yahoo Finance (Yahoo Finance).

In parallel, we construct a corpus of about 440 political speeches and public statements, focusing on communications by Donald J. Trump during his presidency. These texts are time-aligned with financial data to ensure temporal consistency.

From the textual corpus, we extract structured features using natural language processing techniques, including sentiment scores, topic distributions, emotional intensity, and uncertainty-related linguistic indicators. A further step that is proposed but not implemented in this study is to incorporate semantic embeddings derived from pre-trained language models to capture deeper contextual information.

These features are used to identify and quantify economically relevant signals within the speeches, which can then serve as conditioning variables in downstream models. In this study, only topic scores from Seeded LDA were used as predictors due to time constraints. These topic scores are values from 0 to 1 where the sum of all topic scores for a specific transcript is equal to 1.

The final dataset consists of a synchronized time series in which each observation includes both financial indicators and corresponding speech-derived features, enabling us to model the impact of political communication on transitions between latent market regimes.

Method

This project investigates whether semantic patterns in political communications are associated with transitions between latent market volatility regimes. The methodology combines Hidden Markov Models (HMMs), natural language processing (NLP), and Bayesian inference to model how political communication features relate to changes in market volatility dynamics.

Financial Time-Series Modeling

Market volatility was modeled using a Hidden Markov Model (HMM), which assumes that observed market behavior is generated by an unobserved latent market

state. These latent states are interpreted as volatility regimes: low volatility, medium volatility, and high volatility market conditions.

The observed financial variable was constructed from daily VIX behavior. To better capture volatility intensity and regime persistence, the analysis used absolute VIX changes rather than signed directional changes: $|\Delta VIX| = |VIX_t - VIX_{t-1}|$.

A Gaussian HMM with $K=3$ latent states was fit to the time series using the Baum-Welch expectation-maximization algorithm implemented in the `hmmlearn` library (`Hmmlearn`). The model assumes that observed volatility values are generated conditionally on the hidden regime: $Y_t | Z_t = k \sim N(\mu_k, \sigma_k^2)$ where Y_t is the observed volatility measure at time t .

The HMM also estimates transition probabilities between latent regimes: $P(Z_t | Z_{t-1})$ which captures regime persistence and switching dynamics over time.

After fitting the model, inferred hidden states were reordered according to their learned emission means so that low volatility is state 0, medium if state 1, and high is state 2.

This allowed the latent regimes to be interpreted economically while still being inferred probabilistically by the HMM rather than manually defined from the observed data.

Political Communication Feature Extraction

Political communication data was collected through web-scraped transcripts of speeches, interviews, and public statements obtained from the Senate Democrats Caucus. (The Senate Democratic Caucus) The transcripts were cleaned and preprocessed using standard NLP procedures, including lowercasing, punctuation removal, stopword filtering, and adding delimiters.

To extract latent semantic structure from the communications, Non-Negative Matrix Factorization (NMF) topic modeling was applied to the transcript corpus. NMF decomposes the document-term matrix into latent topic mixtures, producing a set of continuous topic scores for each communication event. Exact topics were set and extracted using Seeded Latent Dirichlet Allocation (LDA).

Transition Modeling

After inferring latent volatility regimes from the HMM, regime transitions were modeled using multinomial logistic regression. The objective was to estimate whether NLP-derived communication features influenced the probability of transitioning into different volatility regimes.

The transition model estimated $P(Z_{t+1} = k | Z_t, X_t)$ where X_t is the array of NLP predictors. Transition probabilities were extracted using a typical softmax model:

$$P(Z_{t+1} = k | Z_t, X_t) = \frac{\exp(\alpha_k + \beta_k^T X_t)}{\sum_j \exp(\alpha_j + \beta_j^T X_t)}$$

Where alpha represents the baseline transition tendency and beta represents the effect of topic features on transition probabilities. Current hidden-state indicators were also included as predictors to capture regime persistence.

Two transition-model variants were estimated: first, an NLP-only model using topic features alone and, second, a full model including both NLP features and financial control variables such as lagged VIX values, S&P 500 returns, and rolling market volatility measures.

This comparison was used to determine whether political communication features provided explanatory power beyond broader financial-market conditions.

Bayesian Extension

To quantify uncertainty in the estimated topic effects, the multinomial transition model was extended using Bayesian inference in PyMC. (Abril-Pla et al.) Instead of estimating single point-valued coefficients, posterior distributions were inferred for all transition parameters.

Regression coefficients were assigned weakly-informative Gaussian priors: $\beta_k \sim N(0, \sigma_k)$. Posterior inference was performed using a Monte Carlo method with the No-U-Turn Sampler (NUTS). The Bayesian framework allowed estimation of posterior means highest-density credible intervals, posterior probabilities that effects were positive, and uncertainty in volatility-transition dynamics.

This Bayesian extension provided a probabilistic interpretation of the relationship between political communication topics and market volatility regimes rather than relying solely on deterministic point estimates.

Model Evaluation and Interpretation

The analysis evaluated inferred volatility regime persistence, transition probabilities between regimes, posterior distributions of NLP topic effects, and the stability of topic effects after including financial controls.

Particular attention was given to whether communication topics increased the probability of transitioning into the high-volatility regime. Posterior probability comparisons and transition-probability simulations were used to assess the magnitude and uncertainty of these effects.

Results

Hidden Markov Model Regime Discovery

The Gaussian Hidden Markov Model identified three distinct latent volatility regimes from the VIX time series. The inferred states exhibited clear differences in volatility magnitude and market behavior. The high-volatility regime was associated with elevated absolute VIX changes ($\mu_{|\Delta VIX|} = 2.3500$), while the low-volatility regime corresponded to relatively stable market conditions ($\mu_{|\Delta VIX|} = 0.1855$).

Visual inspection of the inferred state sequence showed that volatility regimes clustered over time rather than alternating randomly. This behavior is consistent with well-known financial stylized facts such as volatility persistence and volatility clustering. (Figure 1)

The learned transition matrix also demonstrated strong regime persistence. In particular low-volatility states were highly likely to remain low-volatility on the following day, high-volatility states exhibited substantial persistence, and direct transitions between extreme regimes were relatively uncommon. (Figure 2)

These findings indicate that the HMM successfully captured meaningful latent market structure rather than arbitrary partitioning of the observed data.

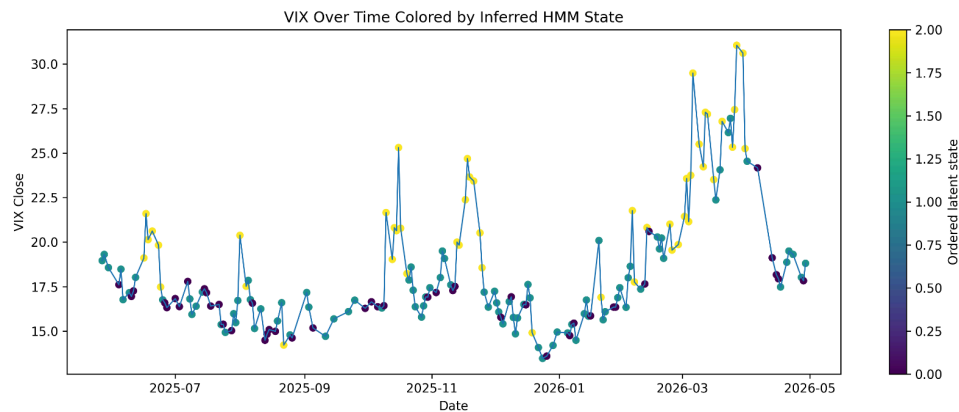


Figure 1. Yellow points indicate high-volatility states, turquoise points indicate medium-volatility states, and purple points represent low-volatility states. The graph tracks the VIX close values over the last 12 months (May 2025 to May 2026) and assigns the HMM-inferred state to each data point.

	To State 0	To State 1	To State 2
From State 0	0.355261	0.556389	0.088349
From State 1	0.257912	0.592653	0.149435
From State 2	0.038180	0.278368	0.683453

Figure 2. The transition matrix showing the probabilities of moving from one state (rows) to another state (columns).

Logistic Transition Modeling

After inferring latent volatility states, multinomial logistic regression was used to model transitions between regimes as a function of political communication topic features.

The baseline transition models included the current volatility regime as a predictor. Across all specifications, the current state variables were consistently the strongest predictors of the next-day state. This result aligns with the Markov assumption underlying the model and reflects the strong persistence structure present in financial volatility regimes.

The NLP-only transition model suggested that several NMF-derived topic features were associated with increased probability of transitioning into the high-volatility regime. The strongest positive effects included:

Variable	Topic Name	Relative Effect
nmf_topic_9_score	Patriotism	Strong positive
nmf_topic_7_score	Media Attacks	Moderate positive
nmf_topic_0_score	Economy, taxes and tariffs	Moderate positive
nmf_topic_4_score	Environment	Moderate positive

Figure 3. Relative effects of the most significant NMF-derived topics.

When financial control variables were added, however, many of these effects weakened substantially or approached zero. (Figure 4) This indicates that much of the apparent explanatory power of political communication topics overlaps with broader market conditions and existing volatility dynamics.

The comparison between NLP-only and full-control models suggests that political communication may contain some information related to volatility transitions, but these effects are relatively modest compared to the endogenous persistence of financial volatility regimes.

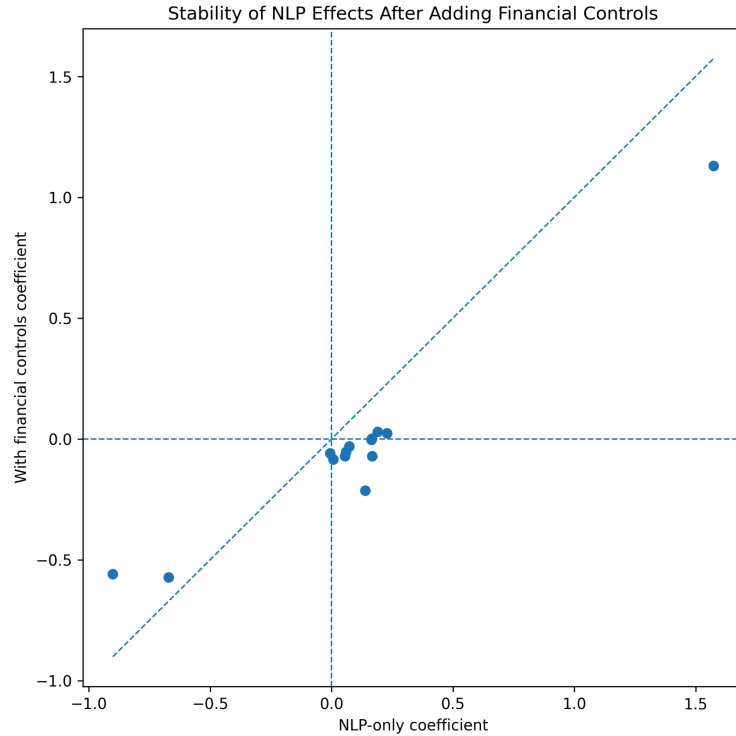


Figure 4. A scatter plot of the values of the coefficients of NLP features in logistic regression before and after adding the financial control variables.

Bayesian Transition Modeling

To quantify uncertainty in the estimated topic effects, the transition model was extended using Bayesian multinomial logistic regression implemented in PyMC. All models converged on all variables with $\hat{r}$ values of at minimum 1.00 and at maximum 1.01.

Posterior inference revealed strong evidence for volatility-regime persistence. The largest and most credible effects were associated with remaining in the current volatility regime, particularly within the high-volatility state. For example, the transition effect of remaining in a high volatility state had strong positive persistence (98.1% probability of a positive effect on transition probability according to HMM trace), while the transition effect of transitioning to a low-volatility state from a high-volatility state was strong and negative.

These findings further support the interpretation that latent market regimes are highly persistent over short time periods.

In contrast, posterior distributions for most NLP topic coefficients were substantially more uncertain. Many topic effects exhibited posterior means in the expected direction, but their highest-density credible intervals often included zero. The results for the same four topics from Figure 3 are below in Figure 5:

Topic	Posterior Mean Effect NLP-Only	HDCI Lower NLP-Only	HDCI Upper NLP-Only	Posterior Mean Effect with Controls	HDCI Lower with Controls	HDCI Upper with Controls
Patriotism	0.099	-0.055	0.285	-0.009	-0.209	0.180
Media Attacks	0.082	-0.075	0.284	-0.006	-0.198	0.193
Economy, taxes and tariffs	0.097	-0.068	0.287	0.014	-0.179	0.213
Environment	0.077	-0.088	0.251	-0.029	-0.215	0.128

Figure 5. Table displaying the posterior mean effects of the four most-significant NMF-derived features and their high-density credible interval bounds both with and without the financial control variables. All values are with regards to transitioning into a high-volatility state.

Although the Bayesian posterior intervals were wide, posterior probability analysis still suggested moderate evidence for positive effects in several topics. Some topic features exhibited posterior probabilities greater than 0.80 that their effects on entering the high-volatility regime were positive.

However, when financial controls were incorporated into the Bayesian model, these topic effects weakened substantially. In many cases posterior means shrank toward zero, credible intervals widened around zero, and even the highest posterior probabilities of positive effect decreased toward approximately 0.50.

This suggests that much of the observed association between political communication topics and volatility transitions may be mediated through broader market conditions rather than representing strong independent causal drivers.

Overall Findings

Across both frequentist and Bayesian transition models, the results consistently suggest that:

1. Financial volatility regimes are highly persistent over time.
2. Political communication topics contain some information associated with volatility-regime transitions.
3. The explanatory contribution of communication features is relatively modest compared to broader market dynamics and existing volatility persistence.
4. Much of the apparent NLP signal overlaps with underlying financial-market conditions.

Overall, the results indicate that political communication may contribute incremental information about short-term volatility-regime dynamics, but market persistence and broader financial conditions remain the dominant drivers of volatility-state transitions.

Citations

Matej Marinč, Nadia Massoud, Riste Ichev, Aljoša Valentinčič, “Presidential candidates linguistic tone: The impact on the financial markets”, *Economics Letters*, Volume 204, 2021, 109876, ISSN 0165-1765, <https://doi.org/10.1016/j.econlet.2021.109876>. (<https://www.sciencedirect.com/science/article/pii/S0165176521001531>)

“VIX Volatility Products.” *Cboe*, www.cboe.com/tradable-products/vix/. Accessed 21 May 2026.

“Trump Transcripts: Senate Democratic Leadership.” Trump Transcripts | The Senate Democratic Caucus, www.democrats.senate.gov/newsroom/trump-transcripts. Accessed 21 May 2026.

“Historical Data for S&P 500 (^GSPC).” Yahoo Finance, Yahoo, <https://finance.yahoo.com/quote/%5EGSPC/history/>. Accessed 21 May 2026.

“HMMlearn Documentation.” Hmmlearn, <https://hmmlearn.readthedocs.io/en/latest/>. Accessed 21 May 2026.

“PyMC: A Modern and Comprehensive Probabilistic Programming Framework in Python”, Oriol Abril-Pla and Virgile Andreani and Colin Carroll and Larry Dong and Christopher J. Fonnesbeck and Maxim Kochurov and Ravin Kumar and Junpeng Lao and Christian C. Luhmann and Osvaldo A. Martin and Michael Osthege and Ricardo Vieira and Thomas Wiecki and Robert Zinkov, *PeerJ Computer Science* Vol. 9 No. e1516, DOI 10.7717/peerj-cs.1516, 2023

Xu, Muxi. “NLP for Stock Market Prediction with Reddit Data.” Department of Computer Science, Stanford University.
https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1214/reports/final_reports/report030.pdf